

Vortex-induced vibration characteristic in two regions of double deck plate-truss composite girder and countermeasure study

Ding Zeng^a, Cunming Ma^{a,b,*}, Cheng Pei^{c,d,e}, Yangzhou Xu^a

^a School of Civil Engineering, Southwest Jiaotong University, Chengdu 610031, China

^b Key Laboratory for Wind Engineering of Sichuan Province, Chengdu 610031, China

^c Airport College, Civil Aviation Flight University of China, Guanghan 618307, China

^d State Key Laboratory of Mechanical Behavior and System Safety of Traffic Engineering Structures, Shijiazhuang Tiedao University, Shijiazhuang 050043, China

^e Sichuan Engineering Research Center for Smart Operation and Maintenance of Civil Aviation Airports, Guanghan 618300, China

ARTICLE INFO

Keywords:

VIV
Plate-truss composite girder
Wind tunnel test
CFD
Fluid-structure interaction
Countermeasure study

ABSTRACT

This research investigates the mechanisms and aerodynamic countermeasures associated with the two vertical vortex-induced vibration (VIV) regions in a double deck plate-truss composite girder. Initially, large-scale wind tunnel tests were conducted on the original section of the girder. These tests revealed the existence of two vertical VIV regions at angles of attack (AoAs) of 0° and $+3^\circ$. Additionally, the computational fluid dynamics (CFD) method was employed to analyze the flow field and simulate the interaction between the flow and structure of the original section. The numerical analysis results suggest that the two VIV phenomena arise from different underlying mechanisms. The first vertical VIV is associated with vortex shedding at the upper chord bar, while the second is related to alternating vortex shedding in two rows downstream of the girder. Finally, based on CFD simulations, aerodynamic countermeasures for VIV were explored, and six different schemes were tested in sectional model wind tunnel experiments. The results demonstrate that the proposed inclined deflector effectively mitigates VIV in both regions of the double deck plate-truss composite girder.

1. Introduction

In the context of long-span bridges, vortex-induced vibration (VIV) represents a significant area of focus within structural wind engineering. The limited magnitude of VIV can adversely affect both the comfort and safety of the driving experience [9], and may lead to localized structural damage, consequently reducing the operational lifespan of various bridge components [10,12]. Truss girders are extensively employed in long-span bridges due to their exceptional rigidity and rapid assembly capabilities. Typically, double deck truss girders are employed to fulfill the requirements of both highways and railways. However, it is generally not allowed for railway bridges. Thus, it is crucial to investigate the VIV characteristics of truss girders. Numerous researchers have explored the vortex-induced vibration behavior of various bridge section types, including twin-box girder sections [2,6] and Π -type composited girders [15,17]. The aerodynamic interactions resulting from the individual components of truss girders are intricate. The flow interacting with the bluff body can generate multiple vortices on the deck, with varying sizes and shedding frequencies. This complexity further influences the VIV

performance of plate-truss composite girders. Recent studies have indicated that the upstream wake flow can inhibit vortex shedding from downstream members [19], while Fang identified the presence of VIV in double-deck truss girders and characterized the associated vortex behavior through computational fluid dynamics (CFD) simulations [8].

VIV is caused by the periodic vortex forces generated by the vortex shedding, which occurs at frequencies that closely align with the natural frequency of the structure. A broad and continuous bluff body typically generates multiple points of separation that vary according to the Strouhal number. The occurrence of vortex shedding, in conjunction with variations in the Strouhal number and the frequency doubling, leads to the manifestation of two distinct locking intervals in the VIV. Diana et al. [5] found two vertical locking regions of VIV for a multi-box girder section in wind tunnel tests on a suspension bridge spanning the Strait of Messina. Hu et al. [11] observed multiple wind locking regions in 1:20 scale wind tunnel tests of streamlined box girder, which corresponded to three vertical vortex-induced vibrations at a $+3^\circ$ wind angle of attack. Noguchi et al. [16] identified two distinct "lock-in" regions by conducting wind tunnel tests on a model of a box girder section that was

* Corresponding author at: School of Civil Engineering, Southwest Jiaotong University, Chengdu 610031, China.

E-mail address: mcm@swjtu.edu.cn (C. Ma).

<https://doi.org/10.1016/j.istruc.2024.107946>

Received 23 May 2024; Received in revised form 25 October 2024; Accepted 27 November 2024

Available online 10 December 2024

2352-0124/© 2024 Institution of Structural Engineers. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

supported by springs and allowed to vibrate freely. Li et al. [13] focused on the mechanism of the two vertical vibration regions of a Π -type girder and reveals the aerodynamic mechanism of vibration reduction measures. Furthermore, Duan et al. [7] and Bai et al. [4] reported the presence of multiple several VIV regions in the wind tunnel tests conducted on open girders and twin-box girders.

Installing of aerodynamic measures is a common and efficient way to reduce VIV in bridges. Vaz et al. [20] investigated the VIV characteristics of a single-cell rectangular box girder at three different angles of attack (AoAs), proposing modifications such as the border beams and rounding the edges of the box girder base to suppress vibrations. Zhou et al. [22] installed vertical stabilization plates on the upper and lower surfaces of the deck to reduce the aerodynamic instability caused by flow. With the advancement of computational simulation technology, numerical simulations based on CFD have become widely used in wind engineering. The combination of CFD simulation and wind tunnel test is particularly useful for investigating the VIV characteristics of bridges and developing mitigation measures [21]. Álvarez et al. [3] conducted a 3D LES simulation to analyze the aerodynamic behavior of the flow field around both stationary and vertically free-to-oscillate rectangular cylinder with 4:1 aspect ratio. Alonso-Estébanez et al. [1] used CFD simulations to determine aerodynamic coefficients related to side force, lift force, and rollover moment for three types of bridge decks: box, girder, and board. Li et al. [14] developed a digital twin system to accurately predict the VIV response of flat, closed-box steel girders by integrating CFD simulations and wind tunnel tests.

For the double deck plate-truss composite girder, previous studies have identified two “lock-in” regions. However, the differences and similarities in the mechanisms generating these two VIV regions have not yet been fully analyzed. Additionally, appropriate vibration reduction strategies need to be proposed for this type of girder, along with an investigation into the mechanisms driving the reduction. In this study, we focus on the cross-section of a double deck plate-truss composite girder exhibiting two vertical VIV regions. The vortex characteristics at different AoAs, along with the corresponding vibration reduction mechanisms, are examined. First, the VIV response of the original section is assessed through wind tunnel testing. Then, variations in the flow field at different AoAs are analyzed using CFD. The underlying cause of the two vertical VIV phenomena is clarified using fluid-structure interaction technology. Based on these results, mitigation measures are proposed, and the mechanisms for suppressing the two vertical VIVs are explained.

2. Wind tunnel test of double deck plate-truss composite girder

2.1. Background of tested section

The segment model under study is designed as a cable-stayed bridge with an overall length of 1260 m. The main span of the bridge extends to 600 m. The layout of the bridge is illustrated in Fig. 1. Due to its traffic demand, the double deck plate-truss composite girder is adopted in the form of the main girder. The upper deck is specifically engineered for vehicular traffic, while the lower deck is exclusively allocated for railway transit. The widths of the upper and lower decks are 34.5 m and 12.8 m respectively, and the height of the truss is 12.8 m. Further details

are provided in Fig. 2.

2.2. Experimental setup

During wind tunnel testing, we employ large-scale model testing to compensate for the inaccuracies resulting from the Reynolds number effect and the limited size of bridge deck support facilities. The spring-supported sectional model is placed in the XNJD-3 wind tunnel, which features a test section measuring 22.5 m in width, 4.5 m in height, and 36.0 m in length. When the velocity of the incoming airflow exceeds 1.5 m/s, the turbulence intensity in the wind tunnel remains below 1 %. To ensure that the model is influenced solely by a two-dimensional flow field, two wooden walls are installed in the center of the tunnel. The model is fixed on a rigid bracket and is suspended between the wooden walls using eight springs. The displacement response of the sectional model is assessed by measuring the sampling points fixed on the rigid support with a laser sensor.

The model has a scaling ratio of 1:23.5. Based on the dimensions of the actual bridge, the model's height is 0.544 m, width is 1.468 m, and its length is 3.9 m. The model's frequency and damping characteristics were determined through free attenuation vibration, which was conducted with excitation at a wind speed of zero. The intrinsic frequencies and damping were determined by averaging the values at three angles of attack (AoAs): 0 and $\pm 3^\circ$. The main parameters of the segment model in the wind tunnel, such as geometric size and mass, are shown in Table 1. Since torsional VIV was not observed during the experiments, the structural intrinsic vibration characteristics for torsional vibration are not presented. The diagram of the model and supporting structure system are shown in Fig. 3, and the model in the wind tunnel is shown in Fig. 4.

2.3. VIV response of the original section

The VIV performance was evaluated at AoAs of 0° and $\pm 3^\circ$ within a uniform flow wind tunnel test, utilizing the original section for testing. Fig. 5 illustrates the outcomes of the tests. In the figure, the wind speed and vibration amplitude have been normalized through dimensionless processing, where U denotes the test wind speed, and B and D represent the characteristic width and height of the model. Two distinct vertical

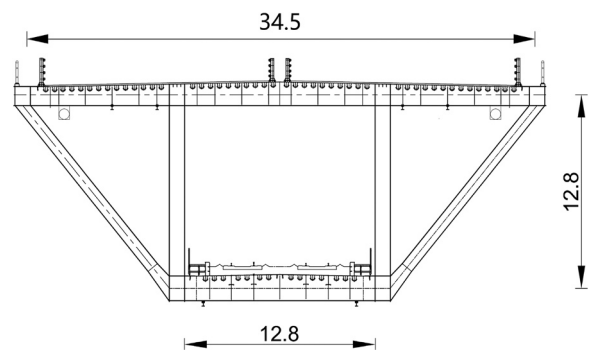


Fig. 2. Section view of the girder (unit:m).

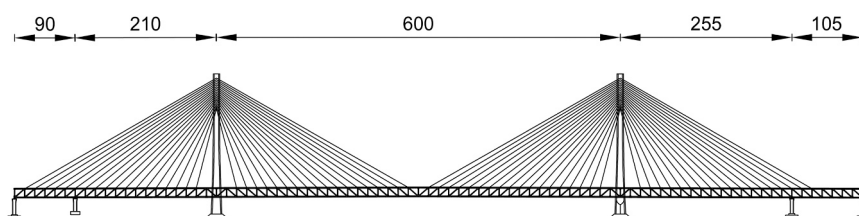


Fig. 1. Layout of the cable-stayed bridge (unit:m).

Table 1

The parameters of scaled sectional model.

Parameter	Unit	Scaling	Prototype	Model
Height (D)	m	1:23.5	12.8	0.544
Width (B)	m	1:23.5	34.5	1.468
Mass (M)	kg/m	1:23.52	57,946	104.4
Vertical Frequency (f_h)	Hz	-	-	2.209
Vertical Damping (ξ_h)	%	-	-	0.22

VIVs were identified at both the 0° and $+3^\circ$ AoAs, whereas no VIV was observed at the -3° AoA. At 0° AoA, the two vertical lock-in were identified within the ranges of 0.64–0.77 and 0.91–1.05, and the maximum of the reduced amplitude (y^*) are 0.0742 and 0.1072, corresponding to reduced wind speeds of $U_r = 0.74$ and $U_r = 0.97$, respectively. At $+3^\circ$ AoA, the first vertical VIV corresponds to the reduced wind speed lock-in range of 0.64–0.77, achieving a maximum reduced amplitude of 0.0859 occurs at $U_r = 0.74$. The second vertical VIV corresponds to a reduced wind speed lock-in range of 0.95–1.28, with a maximum reduced amplitude of 0.1123 occurring at $U_r = 1.12$. Overall, it appears that the VIV performance of the original section is not significantly different at AoAs of 0° and $+3^\circ$, and both performances are inferior to those at -3° .

To gain a deeper understanding of the vibration characteristics of the model during testing, the FFT was utilized to evaluate the temporal progression of the VIV displacement. In order to mitigate spectral aliasing during spectral analysis, the sampled signals are filtered by modifying the cutoff frequency of the filter. The cutoff frequency is set to 16 Hz to isolate the signals of the frequency bands that require analysis. Fig. 6 demonstrates the temporal course of laser displacement and the spectral analysis of the maximum amplitude corresponding to the speeds of $U_r = 0.74$ and $U_r = 1.12$ for the two lock-in intervals at a $+3^\circ$ AoA. The dominant frequency of the vibration was identified as 2.2188 Hz, which is in close to the vertical frequency of the experimental model.

3. Mechanism of VIV in two “lock-in” regions of the original section

3.1. Numerical simulation setup

Aiming to analyze a plate-truss composite girder, a simplified two-dimensional CFD model has been developed. The model ensures that each component has the same cross-section, the total wind force area is equal, and the aerodynamic interference of the horizontal component is similar, so as to simulate the three-dimensional aerodynamic characteristics of the section more accurately [18]. The 2D CFD numerical

simulation of the test section is executed utilizing ANSYS Fluent software. The dimensions of the flow field are established at 30B in length and 30D in width. The computational grid and computational domain are shown in Fig. 7. The grid is a hybrid grid consisting of a structured grid encompassing the outside region and unstructured grids surrounding the model. The left edge of the computational domain is designated as a uniform velocity inlet, with a turbulence intensity of



Fig. 4. The model in the wind tunnel.

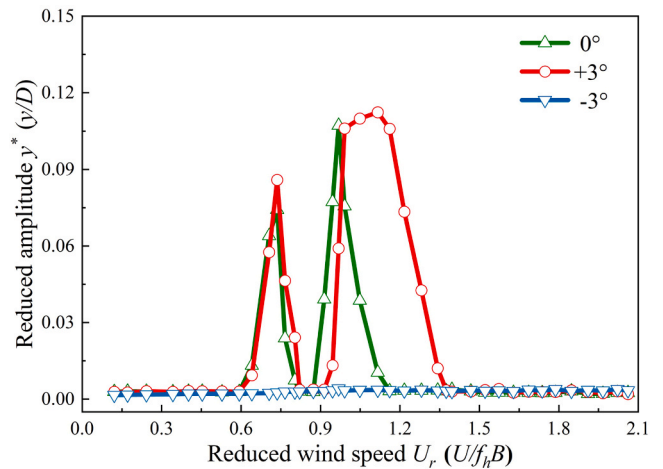


Fig. 5. Results of VIV of original section in wind tunnel tests.

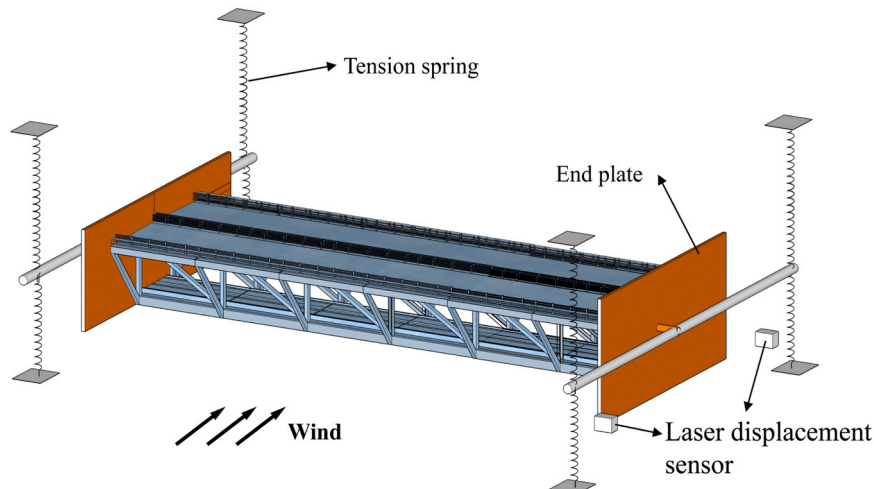


Fig. 3. Schematic diagram of spring support system.

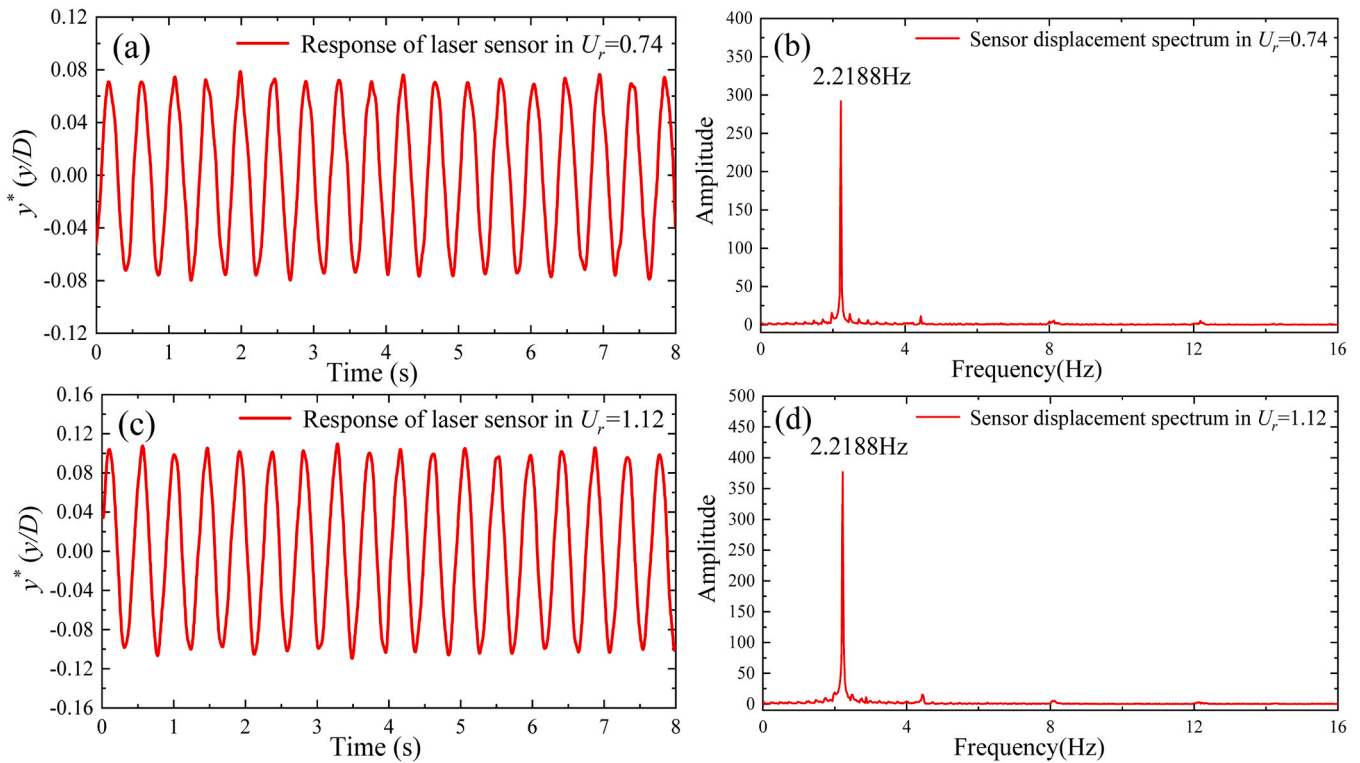


Fig. 6. Time series and amplitude spectrum of displacement under two wind speeds at +3° AoA.

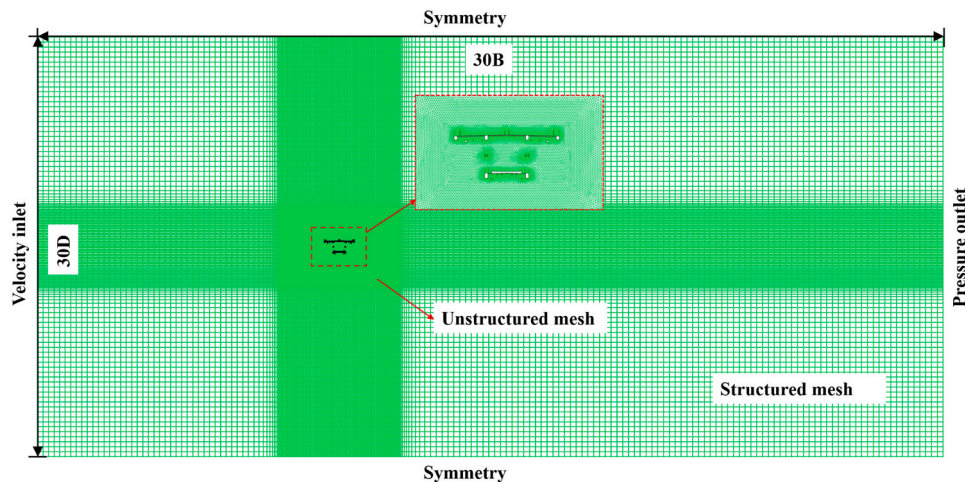


Fig. 7. Computational domain and mesh in the CFD simulation.

0.005 and a turbulence viscosity ratio of 2 are applied at this velocity inlet. The right boundary of the computational domain is designated as a pressure outlet, characterized by a gauge pressure of zero. The upper and lower boundaries are established as symmetry boundaries, while the model is designed with a no-slip wall boundary condition. The number of grid layers in the boundary layer is 15–20 layers. The computation utilizes the SST $k-\omega$ turbulence model, while the solution method employs the SIMPLEC second-order implicit algorithm. The moment equations and turbulence quantities are solved using a second-order upwind scheme, while the pressure equations are calculated using a second-order method during spatial discretization.

3.2. Grid and time-step independence

The flow separation around the truss girder is a critical factor

affecting the vibration caused by the vortex. The simulation of the mesh near the wall plays a crucial role in influencing the flow dynamics of the shear layer. An excessively large time step may obscure the smaller, unsteady flow characteristics. Consequently, it is imperative to mitigate the effects of grid resolution and time step size on the model's accuracy to ensure that the current mesh is suitable for the numerical simulation of plate-truss composite girders. Initially, at the same height of the first grid, different time steps were verified, considering that the time steps are 1×10^{-3} s, 5×10^{-4} s, 1×10^{-4} s and 5×10^{-5} s, with the AoA maintained at 0° . The Reynolds number is 1.3×10^5 . For each scenario, the data were collected over more than 20 cycles after 20 s of calculation, with the incoming wind velocity established at 3.5 m/s, which falls within the lock-in region. The comparison results are displayed in Table 2. It is evident that for the vertical VIV, the aerodynamic lift has the most significant impact, and the lift coefficient of the section at

0° AoA obtained in the test is 0.087. To observe the variation in the lift coefficient under different scenarios, the subsequent simulation time step is set to 5×10^{-4} s. The first grid height of the boundary layer is calculated at a time step of 5×10^{-4} s, and the results are presented in Table 3. Balancing the computational efficiency and accuracy of the results, the height of the first layer was determined to be 2×10^{-5} m for further calculations.

3.3. Comparison of time-averaged flow structures at different AoAs

To investigate the cause of VIV in the girder and analyze its characteristics, static CFD simulations were performed for the original sections at three different AoAs. The CFD setup remains consistent across all AoA conditions, with the computational model and mesh configuration identical to those outlined in Section 3.2. The inlet incoming wind speed was set at 3.5 m/s, which falls within the lock-in interval. The time-averaged flow fields obtained from the simulation are shown in Fig. 8. Vortices of varying scale were generated at different positions around the girder for each AoA. For the flow field around the upper bridge deck, similarities were between the AoAs of 0° and +3°. In both cases, two vortices formed above the upper bridge deck within the designated red box in the figure, while a large one developed below. At an AoA of -3°, no vortex formed above the upper bridge deck, though the flow field below was similar to other AoAs. The flow field distribution around the lower bridge deck showed consistent patterns across all AoAs, with two large vortices present above and below each lower bridge deck. Upon examining the flow field for all three cases, it was found that vortices formed in the upper layer of the girder section at AoAs of 0° and +3°, with large vortices at AoA of +3°. The vortex shedding characteristics under the most unfavorable condition, i.e., AoA of +3°, are further analyzed in the next section by fluid-structure interaction analysis.

3.4. Fluid-structure interaction of original section

The periodic motion of the girder influences the airflow surrounding it, resulting in alterations to the boundary between the fluid and the structure, which subsequently impacts the formation and detachment of vortices. To examine the interaction between the fluid and the girder during the VIV process, the dynamic mesh technique is employed to analyze the fluid-structure interaction of the girder section at +3° AoA, building upon the prior CFD setup. Fig. 9 illustrates the configuration of the dynamic grid employed for the convective field. A rigid region is established around the girder, while a specific range of meshes beyond this region is designated as the deformation domain. The remaining meshes are kept stationary throughout the simulation. During the calculation, the model is initially allowed to perform the calculation in a stationary state. Once the flow field is stabilized, the mesh is updated using the user-defined function (UDF). The girder is then moved under the influence of aerodynamic force, and the vibration time course curve of the girder is recorded. In the UDF, the girder model is simplified to a single degree of freedom (SDOF) system, and the equation of the damped SDOF vibration system are shown in Eq. (1). The SDOF vibration system has the same mass and stiffness as the girder in the wind tunnel test, with the damping ratio set at 0.2 %. The equation is solved using the Newmark-β method.

$$m\ddot{y}(t) + c\dot{y}(t) + ky(t) = F_y(t) \quad (1)$$

Table 2
Simulation results with four time-step schemes.

Time-step (s)	C _L
1×10^{-3}	0.093
5×10^{-4}	0.081
1×10^{-4}	0.086
5×10^{-5}	0.083

Table 3
Simulation results with three first layer height schemes.

First layer height (m)	Gird number	Maximum y^+	C _L
5×10^{-5}	3.14×10^5	2.1	0.081
2×10^{-5}	3.63×10^5	0.88	0.085
1×10^{-5}	3.94×10^5	0.42	0.087

where m , c and k are mass, damping and stiffness of the girder, respectively. $\ddot{y}(t)$, $\dot{y}(t)$ and $y(t)$ are acceleration, velocity and vertical displacement of the girder section, respectively. $F_y(t)$ is the VIF on the girder.

Fluid-structure interaction simulations are performed at different wind velocities, encompassing two regions of VIV. The Reynolds number in the CFD simulations is contingent upon the wind speed, with a range from 3.9×10^4 to 2.4×10^5 . The near-wall mesh characteristics are optimized for different wind speeds, based on the initial mesh configuration. Fig. 10 displays the maximum y^+ values associated with the computational model at various wind speeds, with the highest simulated reduced wind speed yielding a maximum y^+ value of 1.88. Variations in the Reynolds number due to changes in wind speed exert a negligible influence on the grid boundary layer.

Fig. 11 illustrates the outcomes from wind tunnel tests and fluid-structure interaction research conducted at various reduced wind velocities. The CFD results for the first lock-in region from 0.56 to 0.71, which exhibits a slight deviation from the experimental results of 0.64 to 0.82, and the maximum amplitude with the $U_r = 0.66$ is basically the same. In the second lock-in zone, the reduced wind speed region is identical. The CFD result of maximum amplitude is 0.1282 with an U_r of 1.13, which is somewhat greater than the test result of 0.1123. This discrepancy of 12.4 % may be attributed to the damping of the model system during testing. Overall, the CFD results demonstrate a strong correlation with the experimental data.

3.5. Distribution of fluctuating pressure coefficient of the girder

A pressure distribution analysis is conducted at various lock-in intervals to further explore the mechanism of VIV of the main girder. To calculate the surface pressure surrounding the girder, survey the grid cells located along the girder wall and arrange the monitoring points in CFD simulation. Fig. 12 displays the configuration of these monitoring points, which are positioned at specific intervals on both the upper and lower bridge decks as well as along the chords. The arrangement of the monitoring points at the bottom of the orthotropic plate is simplified by using the bottom of the U-rib as a reference surface. One monitoring point is established at the bottom of each U-rib. The data collected from these measurement points encompasses a total of 10 cycles, once the amplitude has reached a stable state in each of the two intervals.

The fluctuating pressure during the VIV of the double deck plate-truss composite girder is investigated, as the average pressure alone impacts the average aerodynamic characteristics and has minimal impact on the VIV. Fig. 13 illustrates the fluctuating pressure coefficient distribution at each monitoring point in the two locking intervals of the girder to enable the analysis of vortex movement and separation on its surface. The analysis is conducted on the fluctuating pressure distributions in the upper and lower decks at wind speeds that correspond to the greatest amplitude in two locking intervals. In both scenarios, a similar pattern of pressure variation is noted on the lower deck of the girder, while the fluctuations in the pressure coefficient on the upper deck surface manifest in distinct regions. Notably, the curve during the first lock-in interval exhibits a sudden change in the H-J area. Higher fluctuating pressure on the upper surface in the J-A region for the second locked wind speed interval condition. Fig. 14 displays the spectral analysis of the pressure time series at the monitoring point position following the curve mutation. The dominant frequency of the

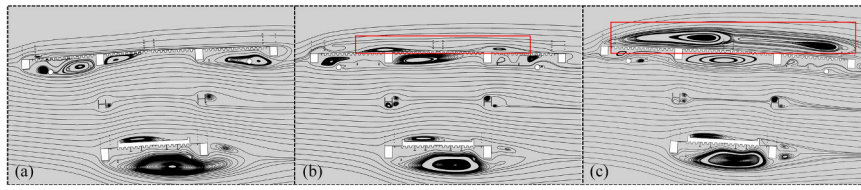


Fig. 8. Time-averaged flow structures of each angles of attack.

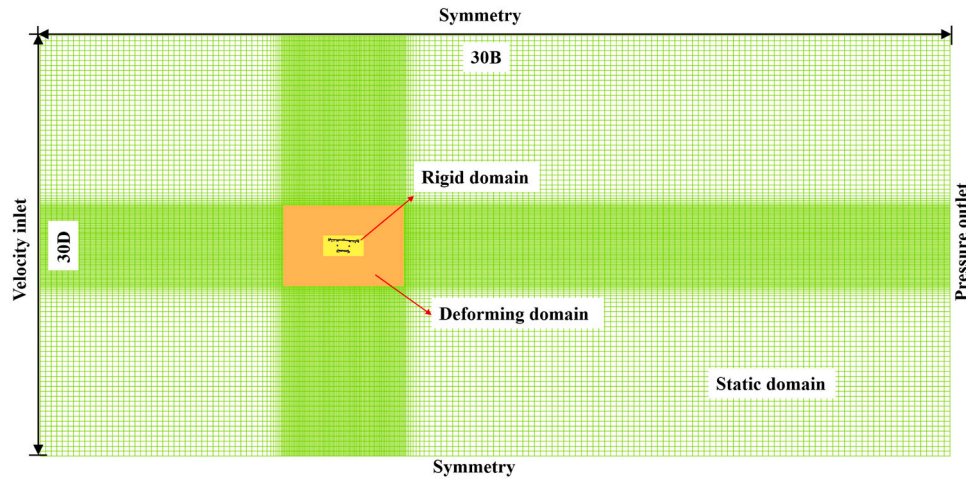


Fig. 9. Grid scheme of fluid-structure interaction analysis.

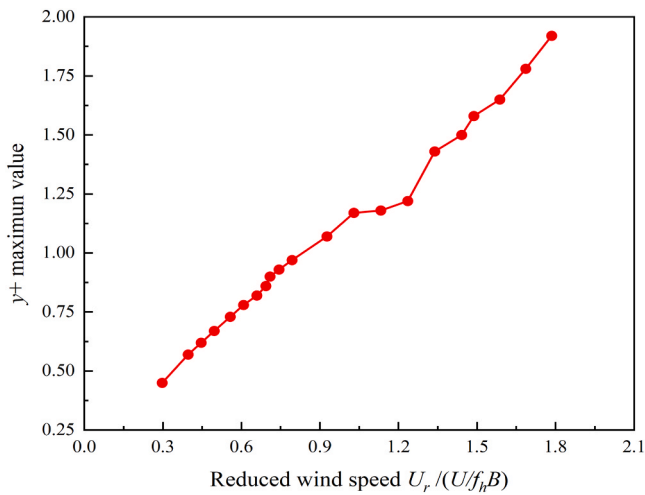


Fig. 10. y^+ Maximum values for each wind speed in CFD simulations.

monitoring point in the H-J region is 2.1934 at $U_r = 0.66$, while the dominant frequency of the pressure at the monitoring point on the upper surface of the girder is 2.1891 at $U_r = 1.13$; both frequencies are comparable to the vortex shedding frequency. The subsequent section will address the flow field properties of the VIV of the girder in relation to the two intervals.

3.6. Characteristics of vortices in the “lock-in” regions

VIV characterization of the girder through the progressive development of vortices in fluid-structure interaction for two wind speed locking intervals. After the amplitude reaches a stable stage, one continuous cycle T of the girder vibration is detected. Fig. 15 shows the evolution of the vorticity in the original section throughout the motion. The two

distinct colors in the figure denote eddies rotating in opposing directions. The expression for the vorticity Z-Vorticity (w_z) can be written as:

$$w_z = \frac{\partial V}{\partial x} - \frac{\partial U}{\partial y} \quad (2)$$

where x and y are the coordinates along the inflow and cross-flow directions, respectively.

As airflow passes through the girder section, vortices of varying sizes form on both the upper and lower surfaces of the bridge. These vortices, located on the upper and lower sides of the lower girder and the lower side of the upper girder, remain attached to the girder and stay relatively stationary during movement. Over time, the vortex above the upper deck interferes with the vortex on the lower side. As vortex Z2 approaches complete detachment from the tail of the chord on the leeward side of the upper deck, vortex Z1 attaches to the chord. Once Z2 detaches, its scale diminishes while Z1 gradually enlarges. The scale of Z1 reaches its maximum value at $T/2$, after which it begins to detach from the chord tail. The periodic attachment and detachment of vortices to the leeward side of the upper deck contribute to VIV in the first lock-in interval.

The evolution of the vorticity in the girder over one aerodynamic cycle is shown in Fig. 16. At $T = 0$, vortex U (as marked in the figure) forms and attaches to the right side of the upper bridge deck. As airflow moves past the railing area, it separates, generating two new vortices, U1 and U2, by $T/4$. During interval between the shedding of U1 and U2, the separated vortices gradually move away from the bridge deck, while vortices generated beneath the upper bridge deck also detach. By $T/2$, vortex U2 has separated from the upper bridge deck, and the vortex below the deck has also detached from chord. Vortex U1 then advances toward the railing area. At $3T/4$, Vortex U1 passes through the railing area and detaches from the girder. This shedding process produces two rows of alternately vortices, forming a “Karman vortex street” at the end of the bridge deck. The alternating shedding of these vortices induces the second VIV of the girder, with the vortex height measuring

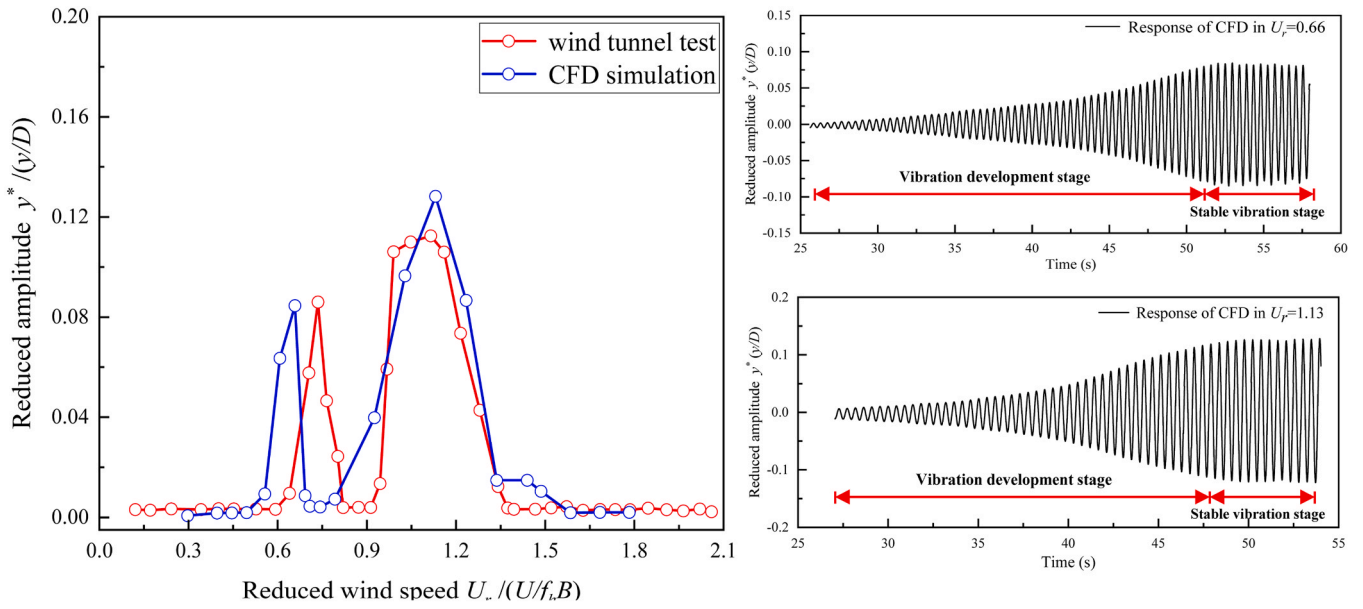


Fig. 11. The original section VIV response obtained by CFD simulation and wind tunnel test at $+3^\circ$ AoA.

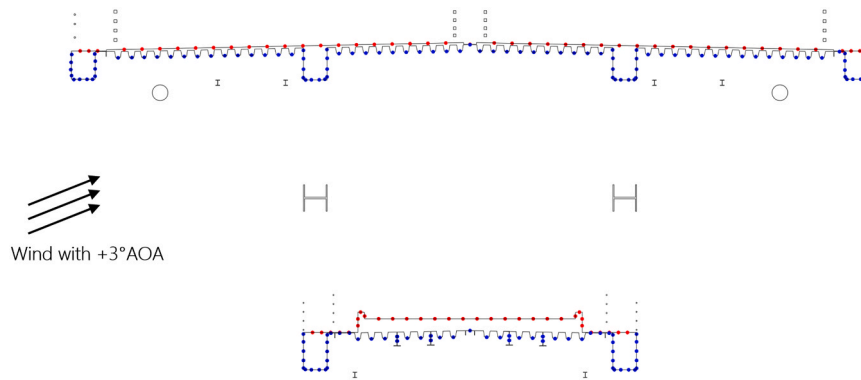


Fig. 12. Distribution of monitoring points in CFD simulation.

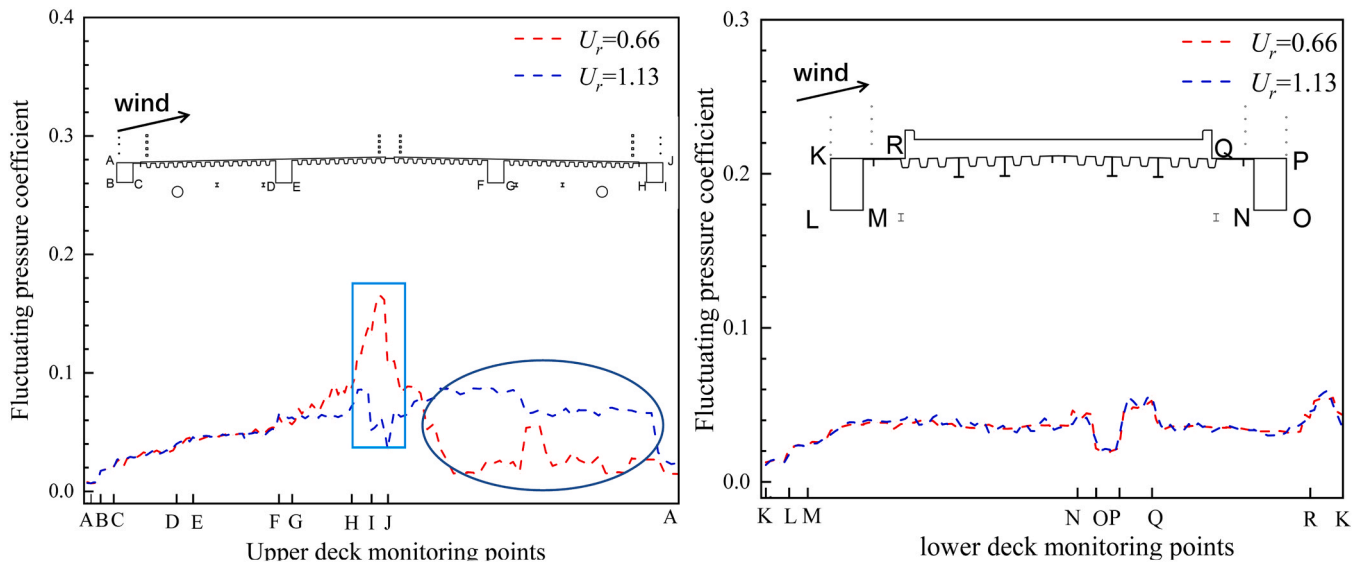


Fig. 13. Distribution of fluctuating pressure coefficient of the girder section for two lock-in regions.

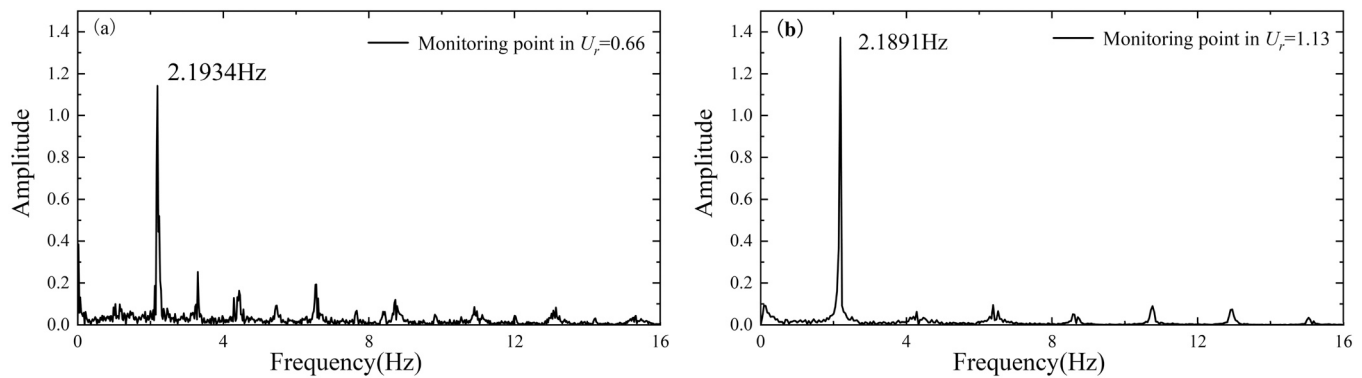


Fig. 14. Spectrum analysis of key monitoring points.

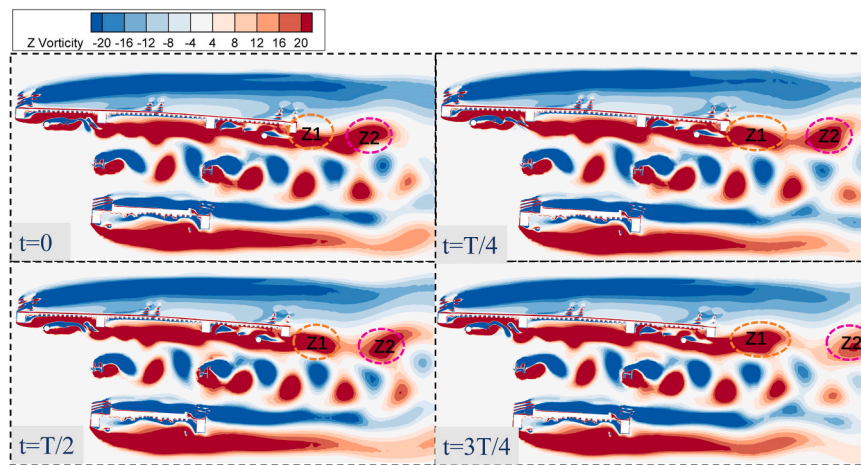


Fig. 15. The development of the flow structures within a vibration period in $U_r = 0.66$.

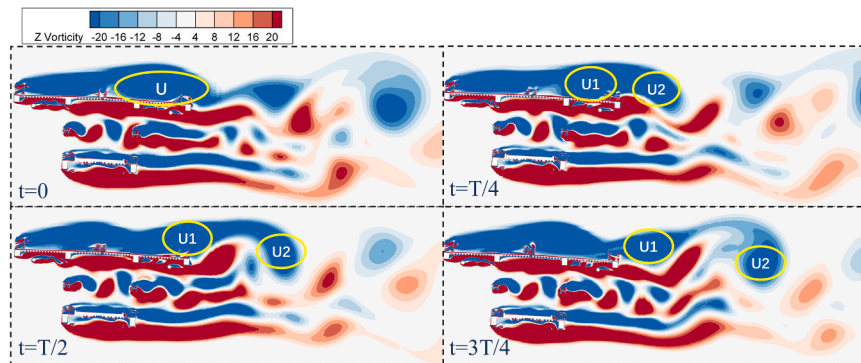


Fig. 16. Vortex evolution diagram within a vibration period in $U_r = 1.13$.

approximately half that of the girder.

The movement of the vortex in the upper deck, as well as the trailing vortex shed via the railing and chord, have the greatest influence on the VIV of the double deck plate-truss composite girder. Distinct absence of vortex shedding is observed in the lower deck, maybe due to its narrower breadth compared to the upper deck in the studied section.

4. VIV control countermeasure study

4.1. Aerodynamic measures in wind tunnel tests

In order to study the suppression of VIV at an AoA of $+3^\circ$, where the amplitude is higher in both regions, several different aerodynamic

control measures were tested in the wind tunnel, as shown in Fig. 17. The results of these tests, involving various aerodynamic devices installed on the girder, are displayed in Fig. 18. Based on the VIV characteristics observed in the two regions of the girder, the aerodynamic measures chosen for the tests were specifically designed to target railing and chord areas of the upper deck. It is important to mention that each aerodynamic measure was implemented simultaneously on both the left and right sides of the girder.

Finding effective aerodynamic measures to control VIV in the wind tunnel is a gradual process. Initially, the study focus on the second lock-in region, where the amplitude is larger due to the different mechanisms of VIV in each region. Within the study of aerodynamic measures, the VIV mechanism of the girder studied in the literature [13] is comparable

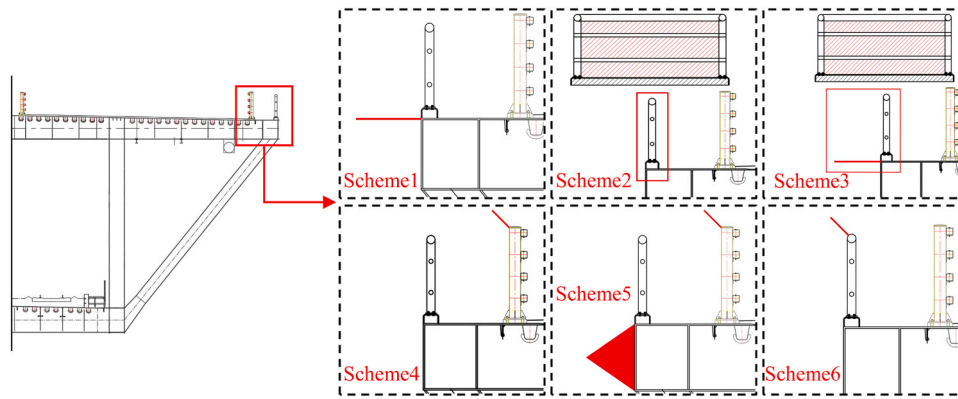


Fig. 17. Schematic diagram of aerodynamic control measures.

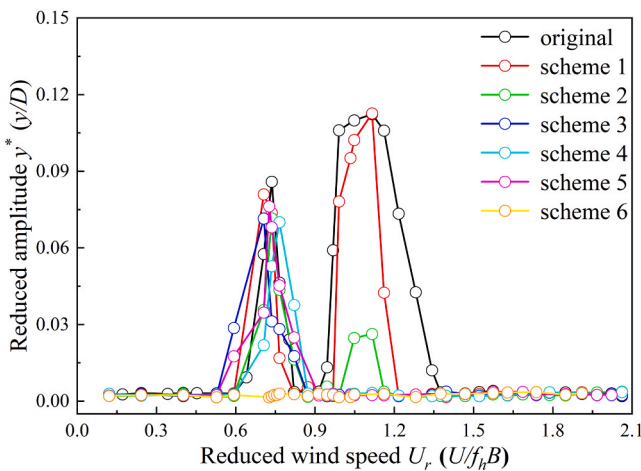


Fig. 18. VIV under different aerodynamic measures in wind tunnel tests.

to the second locking interval of the girder investigated in this research. In that study, a horizontal stabilizer plate was installed to effectively suppress VIV. Hence, the aerodynamic parameter taken into account in the wind tunnel test, Scheme 1 in our test involved placing a 50 mm horizontal stabilizer plate at the same level as the upper deck. However, this measure did not reduce VIV in either region. Next, the test examined the influence of the railing on the VIVs. Scheme 2 evaluated the impact of closing the outermost railing, which effectively suppressed VIV in the second vortex zone, reducing amplitude by 75 %. Scheme 3 combined the features of Schemes 1 and 2. Although this approach reduced vibration in the second region, it had little effect on VIV in the first region.

To simplify the aerodynamic measures, Scheme 4 involved installing a 20 mm long 45° inclined deflector on the guardrails. This measure also effectively reduces vibration in the second lock-in region. Since the VIV in the first lock-in region is caused by vortices formed at the upper chord, Scheme 5 introduced a wind fairing positioned on the outside of the upper chord. However, this arrangement had minimal impact on the VIV in the first region.

A more practical solution was found by integrating aerodynamic measures into the railing area. Scheme 6 involved installing a 20 mm-long, 45° inclined deflector plate on the handrail. The wind tunnel test results indicated that with the implementation of Scheme 6, VIV no longer occurred in the second region and also diminished in the first region. The following section will examine the effectiveness of Scheme 6 in suppressing VIV in both regions, with a particular focus on the mitigation mechanism in the first lock-in region.

4.2. Aerodynamic mitigation mechanism

In order to investigate the mechanism of the suppression of VIV by aerodynamic measures, the evolution of the girder of mounting scheme 6 was investigated using CFD fluid-structure coupling simulations with the airflow at the reduced wind speeds of $U_r = 0.66$ and 1.13. The parameters in the CFD simulation were set the same as in the study of the original section.

Fig. 19 shows the development of the flow field within the girder when Scheme 6 is implemented for one period T at $U_r = 0.66$. In contrast to the original section, the airflow over the upper deck is no longer attached to the chord due to the impact of the deflector plate on the airflow. An analysis of the flow field distribution of the four moments reveals that a portion of the vortex generated by the airflow above the upper deck remains attached to the tail of the deflector plate. Furthermore, the smooth development of the vortex above the upper deck does not impact the vortex on the lower side of the upper deck. Without the influence of the vortex on the upper side, the airflow on the lower side of the upper deck stops to generate attachment as it passes through the chord. Consequently, the VIV in the first interval of the girder section is suppressed.

The development of the flow field around the girder of the installed inclined deflector plates during the second locking wind speed interval is shown in Fig. 20. As the airflow passes through the girder, it ascends diagonally over the deflector plate at the upper deck. After exiting the upper deck, the airflow continues to move for a certain distance and generates a shedding vortex. Concurrently, the airflow at the bottom of the upper deck passes through the chord for a certain distance and generates a shedding effect. An analysis of the four moments in a period reveals that the vortices generated on the top and lower sides of the upper deck no longer interfere with each other, resulting in the absence of alternate shedding vortices. However, both vortices continue to produce shedding at a certain distance from the girder. The distance between the rear shedding vortex and the section is such that it does not

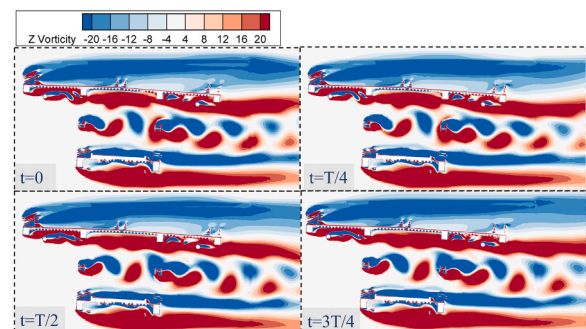


Fig. 19. Flow structures of girder with inclined deflectors in first lock-in region.

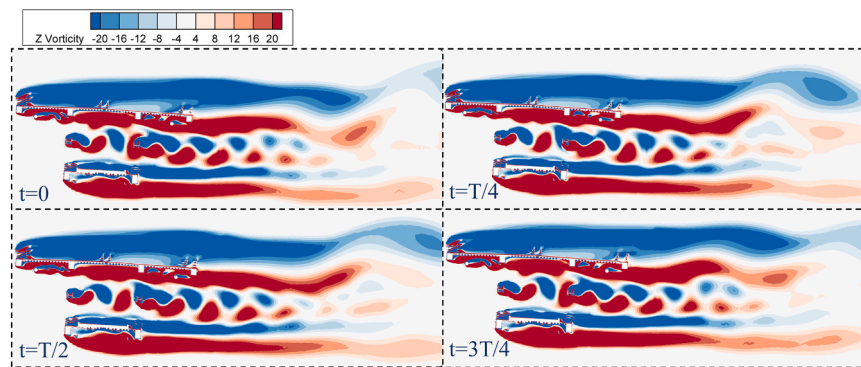


Fig. 20. Flow structures of girder with inclined deflectors in second lock-in region.

impact the girder, resulting in suppression of the girder's VIV.

5. Conclusions

This study investigated the VIV of double deck truss-plate composite girders through wind tunnel testing and CFD simulations. The flow field characteristics of the girder were analyzed in two wind speed locking intervals under vertical VIV conditions. Additionally, aerodynamic measures for mitigating VIV were proposed. The key findings are summarized follows:

- (1) **VIV occurrence and locking intervals:** Vertical VIV in the double deck truss-plate composite girder occurs at AoAs of 0° and $+3^\circ$, with two distinct locking intervals characterized by various amplitudes. However, no VIV is observed at an AoA of -3° . CFD simulations reveal that the variation in AoA leads to vortex formation on the upper deck.
- (2) **Mechanism of the two VIV regions:** Differences in the mechanism of the two vertical VIV regions of the girder. In the first interval, vortex shedding occurs at the end of the upper deck's chord, where airflow passes through the railing, merging with the airflow beneath and forming a vortex behind the chord. This vortex periodically attaches and detaches, causing VIV in the girder. In the second locking interval, the formation of two rows of vortices that alternately shed behind the girder, resulting in VIV.
- (3) **Aerodynamic mitigation measures:** The implementation of a 45° inclined deflector plate on the handrail effectively mitigates VIV in both regions. In the first locking interval, the deflector stabilizes the flow field above the upper deck, preventing periodic vortex shedding on the lower side. In the second locking region, the airflow does not separate after passing through the deflector plate, resulting in the formation of two rows of parallel vortices on both the upper and lower surfaces of the upper deck. The choice of aerodynamic measures and their mitigation mechanisms provide valuable insights for VIV studies of similar girder types.

CRedit authorship contribution statement

Yangzhou Xu: Investigation. **Cheng Pei:** Writing – review & editing, Supervision. **Cunming Ma:** Writing – review & editing, Resources, Funding acquisition. **Ding Zeng:** Writing – original draft, Software, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are grateful for their financial support from the National Natural Science Foundation of China (Grant number 52078438), the Fundamental Research Funds for the Central Universities (PHD2023-024), the Fundamental Research Funds for the Central Universities (CAFUC03037).

References

- [1] Alonso-Estébanez A, Del Coz Díaz JJ, Álvarez Rabanal FP, Pascual-Muñoz P. Numerical simulation of bus aerodynamics on several classes of bridge decks. *Eng Appl Comput Fluid Mech* 2017;11:435–49. <https://doi.org/10.1080/19942060.2016.1201544>.
- [2] Álvarez AJ, Nieto F, Kwok KCS, Hernández S. Aerodynamic performance of twin-box decks: a parametric study on gap width effects based on validated 2D URANS simulations. *J Wind Eng Ind Aerodyn* 2018;182:202–21. <https://doi.org/10.1016/j.jweia.2018.09.004>.
- [3] Álvarez AJ, Nieto F, Nguyen DT, Owen JS, Hernández S. 3D LES simulations of a static and vertically free-to-oscillate 4:1 rectangular cylinder: Effects of the grid resolution. *J Wind Eng Ind Aerodyn* 2019;192:31–44. <https://doi.org/10.1016/j.jweia.2019.06.012>.
- [4] Bai H, Ji N, Xu G, Li J. An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: Sealed traffic barrier. *J Wind Eng Ind Aerodyn* 2020;206:104302. <https://doi.org/10.1016/j.jweia.2020.104302>.
- [5] Diana G, Resta F, Belloli M, Rocchi D. On the vortex shedding forcing on suspension bridge deck. *J Wind Eng Ind Aerodyn*, the eighth Italian national conference on wind engineering in-Vento 2006;94:341–63. <https://doi.org/10.1016/j.jweia.2006.01.017>.
- [6] Duan Q, Ma C, Yang H. Influence of non-uniform grid plates on the vortex-induced vibration performance of twin-box girder section. *Structures* 2023;58:105542. <https://doi.org/10.1016/j.istruc.2023.105542>.
- [7] Duan Q, Ma C, Li J, Li Q, Shang J. Vortex-induced vibration characteristics of two open girders: a comparison of experimental and numerical investigation. *Adv Struct Eng* 2022;25:2844–56. <https://doi.org/10.1177/13694332221101231>.
- [8] Fang C, Hu R, Tang H, Li Y, Wang Z. Experimental and numerical study on vortex-induced vibration of a truss girder with two decks. *Adv Struct Eng* 2021;24:841–55. <https://doi.org/10.1177/1369433220969026>.
- [9] Fujino Y, Yoshida Y. Wind-induced vibration and control of Trans-Tokyo Bay Crossing bridge. *J Struct Eng* 2002;128:1012–25. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2002\)128:8\(1012\)](https://doi.org/10.1061/(ASCE)0733-9445(2002)128:8(1012)).
- [10] Ge Y, Zhao L, Cao J. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2022;220:104866. <https://doi.org/10.1016/j.jweia.2021.104866>.
- [11] Hu C, Zhao L, Ge Y. Mechanism of suppression of vortex-induced vibrations of a streamlined closed-box girder using additional small-scale components. *J Wind Eng Ind Aerodyn* 2019;189:314–31. <https://doi.org/10.1016/j.jweia.2019.04.015>.
- [12] Larsen A. Aerodynamic aspects of the final design of the 1624 m suspension bridge across the Great Belt. *J Wind Eng Ind Aerodyn* 1993;48:261–85. [https://doi.org/10.1016/0167-6105\(93\)90141-A](https://doi.org/10.1016/0167-6105(93)90141-A).
- [13] Li J, Wu P, Hao J, Pan H. Experimental and numerical studies on the two “lock-in” regions characteristic of vertical vortex-induced vibration of H-shaped composite bridge deck. *J Wind Eng Ind Aerodyn* 2022;228:105122. <https://doi.org/10.1016/j.jweia.2022.105122>.
- [14] Li H-Y, Xu Y-L, Wang B, Zhu L-D, Meng X-L, Zhang G-Q. Establishment and application of a digital twin for vortex-induced vibration of a bridge deck section. *Eng Appl Comput Fluid Mech* 2024;18:2297032. <https://doi.org/10.1080/19942060.2023.2297032>.
- [15] Liu Zhiwen, He F, Yan A, Liu Zhenbiao, Yin T, Chen Z. Mechanism and aerodynamic countermeasures of vortex-induced vibration of a cable-stayed bridge

- with narrow I-shaped girder sections. *J Bridge Eng* 2023;28:04023090. <https://doi.org/10.1061/JBENF2.BEENG-6285>.
- [16] Noguchi K, Ito Y, Yagi T. Numerical evaluation of vortex-induced vibration amplitude of a box girder bridge using forced oscillation method. *J Wind Eng Ind Aerodyn* 2020;196:104029. <https://doi.org/10.1016/j.jweia.2019.104029>.
- [17] Owen JS, Vann AM, Davies JP, Blakeborough A. The prototype testing of Kessock Bridge: response to vortex shedding. *J Wind Eng Ind Aerodyn*, the wind engineering society's second UK conference 1996;60:91–108. [https://doi.org/10.1016/0167-6105\(96\)00026-8](https://doi.org/10.1016/0167-6105(96)00026-8).
- [18] Tang H, Li Y, Wang Y, Tao Q. Aerodynamic optimization for flutter performance of steel truss stiffening girder at large angles of attack. *J Wind Eng Ind Aerodyn* 2017;168:260–70. <https://doi.org/10.1016/j.jweia.2017.06.013>.
- [19] Tang H, Shum K, Tao Q, Jiang J. Vortex-induced vibration of a truss girder with high vertical stabilizers. *Adv Struct Eng* 2019;22:948–59. <https://doi.org/10.1177/1369433218778656>.
- [20] Vaz DC, Almeida RAB, Didier E, Urgueira APV, Borges ARJ. Improving the aerodynamic performance of Vila-Real Bridge deck-section. *J Wind Eng Ind Aerodyn* 2016;156:72–83. <https://doi.org/10.1016/j.jweia.2016.07.002>.
- [21] Xu K, Ge Y, Zhao L, Du X. Experimental and numerical study on the dynamic stability of vortex-induced vibration of bridge decks. *Int J Struct Stab Dyn* 2018;18:1850033. <https://doi.org/10.1142/S0219455418500335>.
- [22] Zhou R, Yang Y, Ge Y, Zhang L. Comprehensive evaluation of aerodynamic performance of twin-box girder bridges with vertical stabilizers. *J Wind Eng Ind Aerodyn* 2018;175:317–27. <https://doi.org/10.1016/j.jweia.2018.01.039>.